1 Nature and measurements

01. Palmer Penguins: species and body measurements

Download. Palmer Penguins project. CSV files are available from the project's linked repository, including `penguins.csv`.

The table contains 344 penguins from three species in the Palmer Archipelago, Antarctica. It records bill length and depth, flipper length, body mass, island, sex, and year. The variables have familiar physical meanings, and the whole table is small enough to examine directly.

Several measurements may convey similar information about a bird. Others may help distinguish species even when birds have similar masses. Island adds a different kind of information, since the observed species are distributed unevenly across the sampled locations. A species can also contain substantial variation in body size, so distinguishing species and describing variation within a species are different questions.

Questions. How much uncertainty about species remains when only body mass is known? Does bill shape add information beyond flipper length? How much of the information provided by island is already present in the physical measurements?

02. Mushroom: visible properties and recorded edibility

Download. UCI Mushroom. The download includes a text table and descriptions of its coded categories.

There are 8,124 records with categorical properties such as cap shape, cap colour, odour, gill characteristics, and habitat. Each record is labelled edible or poisonous. The records were constructed from mushroom descriptions in a field guide, rather than collected as a random sample of mushrooms in the wild.

The many categories allow questions about the information carried by individual observations and by combinations of observations. A property can be common without helping distinguish the two labels. Conversely, a relatively simple property may be informative. The table also raises the question of whether several visible properties repeat the same information or distinguish cases that would otherwise look alike.

Questions. Which properties leave the most uncertainty about the recorded label? Can two properties be informative together even when each alone is not? Does habitat contribute information once appearance and odour are known?

03. Wine Quality: chemistry and sensory ratings

Download. UCI Wine Quality. Two CSV files contain red and white Portuguese Vinho Verde wines.

Each wine has measurements including acidity, residual sugar, density, alcohol content, and sulphates, together with a sensory quality score. Red and white wines are supplied separately. The score is an ordered assessment, while the chemical measurements describe different aspects of the same product.

The distinction between chemical composition and perceived quality gives the data a clear source of interest. Wines with similar chemistry need not receive identical scores, and several chemical variables may convey overlapping information. Red and white wines may also have different distributions of the same measurement. This allows questions about both differences between populations and uncertainty within a population.

Questions. How much information about quality is contained in alcohol content or acidity? Do the chemical variables tell us more about wine colour than about quality? Which measurements add information about the score once the others are known?

2 People and public life

04. Adult: work, education, and recorded income

Download. UCI Adult. The archive contains text tables and a variable dictionary.

This historical dataset contains 48,842 records extracted from the 1994 United States census database. Variables include age, education, occupation, working hours, marital status, and other recorded characteristics. The income label distinguishes annual income above or below 50,000 US dollars.

Education, occupation, and working hours describe related aspects of a person's economic circumstances. They may therefore carry overlapping information about the income label. The data also make it possible to ask whether an association seen across all records is equally informative within different groups. The year and the selection of records matter when interpreting what the table represents.

Questions. How much uncertainty about the income label remains after education is known? Does occupation add information beyond education and working hours? Are the same variables equally informative across age groups, or does their relation to the label change?

05. IBGE Census 2022: the age structure of Brazil

Download. IBGE Census 2022 and SIDRA. Population tables by age, sex, and location can be exported as CSV or spreadsheets.

These tables give population counts for Brazil's regions, states, and municipalities. They describe distributions of residents across age groups and other published categories. A municipality's population size and its age composition answer different questions, even though both are recorded in the same collection.

Locations with similar population totals can have different age structures. Distant places can also have closely related distributions. A national age distribution can conceal differences between municipalities, even when it accurately describes the country as a whole. The same population can therefore look more or less varied according to the geographical detail retained in its description.

Questions. How informative is municipality about a resident's age group? Which places have similar age distributions despite different population sizes? How much information about age composition disappears when municipalities are combined into states?

06. TSE Elections 2022: votes and geographical variation

Download. TSE results for 2022. CSV resources include votes by candidate or party, municipality, and electoral zone.

The official results record how votes were distributed across candidates and locations in Brazilian elections. The portal provides separate resources for different offices and levels of geographical detail. The observations are published vote counts, rather than individual voters' complete choices across offices.

Some places distribute votes among many alternatives, while others concentrate them more strongly. Candidates can also have geographically concentrated or widely dispersed support. These differences are closely related to entropy and to the information that location provides about a recorded vote. National totals can conceal substantial local variation.

Questions. Where is the distribution of votes most concentrated or most diverse? How much information does municipality provide about the candidate receiving a vote? How much local variation is lost when results are described only by state or by national totals?

3 Commerce and choices

07. Online Retail: purchases, products, and customers

Download. UCI Online Retail. The data are supplied as an Excel workbook.

The table contains about 542,000 invoice lines from a United Kingdom online retailer in 2010–2011. Fields include invoice identifier, product, quantity, price, date, customer identifier, and country. Several rows can belong to one purchase because an invoice can contain several products.

Products can occur together, customers can return, and demand can change with the calendar. A product that is frequent overall may be unusual for a particular customer or country. Repeated purchases may reveal a stable preference, while a combination of products may reveal something that none of its items conveys alone. The order and grouping of purchases both matter.

Questions. Does knowing one product in a purchase reduce uncertainty about the other products? How informative are previous purchases about a customer's next choice? Does country reveal preferences beyond the products' overall popularity, and how much does the calendar contribute?

08. Olist: Brazilian online purchases and delivery

Download. Olist Brazilian E-Commerce on Kaggle. The release contains several related CSV tables.

The collection covers roughly 100,000 orders from Brazilian marketplaces during 2016–2018. Its tables describe orders, items, products, customers, sellers, payments, delivery dates, and review scores. It brings together the commercial transaction and the customer's later response to it.

An order's price, product category, seller, and delivery history may carry different information about satisfaction. Several of these variables can also reflect the same underlying circumstances. Geography gives the collection a further dimension: delivery experiences and purchasing patterns may differ across locations. The distinction between an order and its individual items is visible in the way the data are organised.

Questions. How much uncertainty about a review score remains when delivery time is known? Does product category add information beyond price? How informative are customer and seller locations about delivery experience, and do their contributions overlap?

09. MovieLens 100K: film ratings and shared tastes

Download. GroupLens MovieLens 100K. A ZIP archive contains ratings, film information, and file descriptions.

This fixed release has 100,000 ratings from 943 users for 1,682 films. Ratings run from one to five, and the film information includes genre categories. Each rating identifies a user and a film, so the same people and films appear repeatedly.

Two viewers may agree on some films and disagree on others. A highly rated film can also receive different reactions from different audiences. The data therefore connect personal taste with uncertainty about an unobserved rating. Genre is a compact description of a film, while a viewer's previous ratings provide a more individual description of preferences.

Questions. How much information does genre provide about a rating? Do earlier ratings by the same viewer reduce uncertainty more than a film's overall popularity does? How much information about a film's reception is lost when its full rating distribution is replaced by its average?

4 Transport and energy

10. Bike Sharing: weather and urban demand

Download. UCI Bike Sharing. The archive includes `hour.csv` and `day.csv`.

The records describe bicycle rentals in the Capital Bikeshare system during 2011–2012. Rental counts are accompanied by calendar and weather variables, including temperature, humidity, and weather categories. The hourly and daily files give two different views of the same system.

Demand has recurring patterns, but similar days need not have the same number of rentals. Weather and the working calendar may carry partly overlapping information. Combining hourly observations into daily totals also hides the timing of demand within a day. Two days with the same total can have different morning and evening patterns, and a familiar pattern can still contain unexpected fluctuations.

Questions. How much do hour and day of the week tell us about demand? Does weather add information once the calendar is known? How much of the next hour's uncertainty is reduced by knowing recent demand, and what is lost in daily totals?

11. New York taxi trips: origins, destinations, and fares

Download. New York City Taxi and Limousine Commission. Monthly trip files are in Parquet format, with a CSV taxi-zone lookup table.

The yellow and green taxi files record pickup and drop-off times and zones, trip distances, and fare-related fields. The available months cover many years, making it possible to look at travel within a particular period or at changes across periods.

An origin does not determine a destination, but some routes recur more often than others. Time of day may change the range of likely destinations and journey durations. Distance, duration, and fare are related descriptions of a trip, although they need not carry identical information. The geographical zones also provide a natural comparison between a detailed description and a broader one.

Questions. How informative are pickup zone and time about destination? Which origins have the greatest uncertainty about where passengers go? Does trip distance leave much uncertainty about fare or duration, and how much route information disappears when neighbouring zones are combined?

12. Open Power System Data: electricity, wind, and sunlight

Download. Open Power System Data, release 2020-10-06. The hourly CSV is `time_series_60min_singleindex.csv`.

This release contains European electricity demand, wind and solar generation, and other electricity-system series for 2015 to mid-2020. The available variables differ across countries and areas. The page also provides files at finer time resolutions where the original observations support them.

Electricity demand and renewable generation combine regular cycles with less predictable variation. Solar and wind generation need not fluctuate together, and a neighbouring area's observations may contain information about a local series. A source that supplies little information on its own may add useful information when another source is already known. The value of geographical and temporal detail can differ between demand and generation.

Questions. How much does the hour or season tell us about electricity demand? Does recent wind generation add information beyond the calendar? How much information does one country's generation provide about another's, and do wind and solar output convey complementary information?

5 Words and opinions

13. SMS Spam: short messages and unwanted advertising

Download. UCI SMS Spam Collection. The archive contains a text file with a label and message on each line.

The collection has about 5,600 English-language messages labelled ordinary or spam. Messages are short, but they can contain distinctive words, punctuation, telephone numbers, and repeated phrases. The label and the original text are both available.

Some messages reveal their character through a few words, while others use language that occurs in both categories. Repeated phrases may make messages easier to compress without necessarily conveying new information about the label. Short texts also make the relation between context and uncertainty particularly visible: a single word may be ambiguous until its neighbours are known.

Questions. Which parts of a message carry information about its label? How much information is retained by the first few words or by message length? Does the combination of words reveal something that their separate frequencies miss, and are more repetitive messages also easier to identify?

14. AG News: the language of different news topics

Download. AG News on Hugging Face. The repository provides downloadable data files and a description of the release.

The dataset contains short news texts labelled world news, sports, business, or science and technology. It has 120,000 training examples and 7,600 test examples. The texts give a relatively compact view of the language associated with four broad subjects.

Some words identify a subject directly, while others occur across many subjects. A company name can occur in business or technology news, and a country name can occur in politics or sport. This overlap makes it interesting to ask how topic information is distributed through a text and how much depends on combinations of words.

Questions. How much uncertainty about topic remains after a short fragment is read? Which words convey similar topic information, and which add something different? How much information about topic survives when names, punctuation, or less frequent words are absent?

15. IMDb reviews: opinion, length, and ambiguity

Download. IMDb on Hugging Face or the original Stanford release. Both identify the Large Movie Review Dataset.

The labelled collection has 25,000 training and 25,000 test reviews, divided between positive and negative opinions. The original release also contains unlabelled reviews. The labelled selection emphasises clearly positive or clearly negative ratings, rather than all possible opinions about films.

Reviews can express an opinion through direct adjectives, narrative detail, or a change of tone. A favourable word can appear inside an unfavourable sentence, especially when negation is involved. Long reviews may provide more evidence while also repeating an opinion. A passage can even praise one aspect of a film and criticise another, leaving its overall judgement uncertain until more of the review is read.

Questions. How much information about the label is contained near the beginning of a review? Does a longer review always reduce uncertainty? What information is lost when word order is ignored, particularly in sentences with negation or mixed praise and criticism?

6 Text and sound

16. WikiText-2: repetition and predictability in written language

Download. WikiText on Hugging Face. The `wikitext-2-raw-v1` configuration provides the small corpus without the processed version's vocabulary substitutions.

WikiText-2 is drawn from English Wikipedia articles. It preserves longer passages of text, so a word can be considered within a sentence, paragraph, or article. The release supplies separate training, validation, and test portions.

The same text contains repetition at several scales. Letters constrain nearby letters, words constrain later words, and an article's subject makes some vocabulary more likely throughout a passage. Yet names and changes of subject can remain difficult to anticipate. A longer context may explain a reference or suggest a word that would be surprising within a single sentence.

Questions. How much less uncertain is the next word when a longer passage is known? Does the article's topic provide information beyond the nearby words? Which parts of the text are most repetitive, and how much information disappears when punctuation, capitalisation, or paragraph boundaries are removed?

17. FMA: music genres and recorded sound

Download. Free Music Archive dataset. The metadata ZIP includes `tracks.csv`, `genres.csv`, and `features.csv`. The separate `fma_small.zip` contains audio.

The small audio collection has 8,000 thirty-second music excerpts in eight genre categories. The metadata release also provides track descriptions and numerical summaries of sound. Those tables can be obtained independently of the audio files.

Genre names are broad descriptions of music, while a recording contains rhythm, timbre, and other details. Similar sounds can occur under different labels, and tracks within one genre can be diverse. A compact sound summary may preserve a distinction between genres while omitting other audible differences. A short excerpt can also suggest a different character from another part of the same track.

Questions. How much uncertainty about genre remains after a sound summary is known? Do rhythm-related and timbre-related descriptions provide different information? How much genre information is present in a short excerpt, and which musical distinctions survive a coarser recording?

18. ESC-50: identifying everyday sounds

Download. ESC-50 repository. The download contains WAV recordings and a CSV metadata table.

The collection contains 2,000 five-second recordings divided equally among 50 sound categories. Examples include animals, rain, engines, footsteps, and household sounds. Each recording has a category label, and the files include information about their source and the supplied partitions.

Different sources can produce acoustically similar sounds. A recording may also contain background noise or a sound whose identity becomes apparent only after several seconds. The label usually names a particular kind of event, but it does not describe everything audible in the clip. A brief sound can be ambiguous even when a longer recording makes its source clear.

Questions. Which categories remain difficult to distinguish from their sound alone? How much information is gained by hearing more of the five-second clip? Can a less detailed recording preserve the event's identity even when it loses other audible features?

7 Images, art, and translation

19. Fashion-MNIST: small images of clothing

Download. Fashion-MNIST repository. The release links to compressed image and label files in IDX format.

There are 60,000 training and 10,000 test images. Each image has 28 by 28 greyscale pixels and one of ten clothing labels, including shirts, trousers, footwear, and bags. The small images are simple enough to view individually while still containing differences in outline, texture, and brightness.

Some clothing categories have similar silhouettes. Other categories may be distinguished by a small region of the image. Pixel values therefore contain both information relevant to the label and detail that may be unnecessary for that particular distinction. Preserving the appearance of a garment and preserving enough information to recognise its category can require different amounts of detail.

Questions. How much information about the label is carried by outline rather than brightness? Which categories become ambiguous when detail is lost? How small can an image description become while still preserving useful information about the type of clothing?

20. MoMA collection: artists, periods, and artistic media

Download. Museum of Modern Art collection repository. `Artworks.csv` and `Artists.csv` are also available as JSON. These files contain metadata, not artwork images.

The artwork records include titles, artists, dates, media, dimensions, and acquisition information. Artist records include biographical fields where available. The collection covers paintings, photographs, prints, design objects, and other forms represented in the museum.

A museum collection reflects changes in artistic production and in acquisition choices. The distribution of media may differ across periods, artists, or acquisition dates. Several labels can also be broad or incomplete descriptions of a work. A collection can contain many objects while drawing heavily on a few media or artists, so the number of works alone says little about its diversity.

Questions. How does the entropy of artistic media differ across periods? How informative are an artist's dates or recorded nationality about medium? How much information about the collection is lost when distinct materials are combined into broad categories?

21. Tatoeba: sentences and their translations

Download. Tatoeba downloads. The "Sentence pairs" export provides sentences in a chosen language with translations in another, including Portuguese and English.

Tatoeba is a collection of contributed sentences and translation links. Its downloads provide sentence texts and language identifiers, with broader exports available for the collection and its links. A sentence can have several translations rather than a single uniquely determined counterpart.

Translations express related content through different sequences of words. Languages can differ in sentence length, word order, and the ways they express negation or grammatical distinctions. Alternative translations also make visible the uncertainty that remains even when the source sentence is known. Several expressions can convey similar content, while a small change of wording can introduce a distinction absent from another translation.

Questions. How informative is the length of a sentence about the length of its translation? Which words or constructions have several common counterparts? What information about word order or grammatical distinctions is preserved across the chosen language pair?

8 Movement and health

22. Human Activity Recognition: movement recorded by a phone

Download. UCI Human Activity Recognition. The ZIP contains sensor signals, numerical summaries, activity labels, and subject identifiers.

Thirty volunteers performed six activities while carrying a smartphone, including walking, sitting, and standing. The release divides the recordings into short windows and supplies summaries of accelerometer and gyroscope signals. Different windows can belong to the same person and activity.

Movements can leave different patterns in the two sensors, while some activities can produce similar signals. A person's way of moving may also be visible independently of the activity. A short window may make an action clear, or it may contain a movement shared by several actions. A numerical summary can preserve some of these distinctions while omitting their order in time.

Questions. How much information about activity comes from acceleration, and how much does rotation add? Which activities remain difficult to distinguish? Does the signal contain information about the person as well as the action, and is activity information equally clear for different people?

23. MIT-BIH: heartbeats and electrical signals

Download. MIT-BIH Arrhythmia Database on PhysioNet. The release contains WFDB signal files, headers, and beat annotations.

The database has 48 half-hour, two-channel electrocardiogram recordings from 47 subjects. The electrical signals are accompanied by annotations identifying beats and their recorded types. The ordered signal and the sequence of labelled beats provide related descriptions of the same recording.

A heartbeat has a shape, but it also occurs within a temporal pattern. Successive beats can resemble one another or differ in timing and form. Two channels observe related electrical activity from different leads. A beat may therefore be less uncertain when neighbouring beats are known, and a feature visible in one channel may be clearer or less apparent in the other.

Questions. How much does the recent sequence of beats tell us about the next one? Does the second channel add information about beat type? How much information is retained by beat timing alone, compared with a description that also includes the signal's shape?

24. World Bank indicators: income and life expectancy

Download. Life expectancy and GDP per capita at purchasing power parity. Each indicator page offers a CSV download.

The tables report country-level indicators over time. Life expectancy describes a population's mortality conditions, while GDP per capita gives an economic measure. Country names, years, and the available numerical values accompany each series.

Countries with similar income can have different life expectancies, and their paths over time can differ. The association between the two indicators may also change across periods or regions. The data give a familiar setting in which to distinguish a broad relation from the uncertainty that remains around it. A regional description is more compact than a country description but may hide meaningful differences.

Questions. How much information does income provide about life expectancy across countries? Does region add information once income is known? Has the uncertainty about life expectancy at similar income levels changed over time, and what differences disappear in regional summaries?

9 Weather, Earth, and space

25. NOAA weather stations: daily temperature and rainfall

Download. NOAA Global Historical Climatology Network Daily. The product page links to station archives and Climate Data Online exports.

The collection contains daily station observations, including minimum and maximum temperature and precipitation. It covers many locations and long periods, with the available dates and variables differing by station. Station information and quality flags accompany the observations.

Weather has recurring seasonal patterns and shorter fluctuations. Rainfall is also different in character from temperature: many days can be dry, followed by episodes of very different intensity. Nearby stations may share information without behaving identically. A monthly average can describe a season while concealing the order of wet and dry days or the occurrence of a few extreme observations.

Questions. How much does season reduce uncertainty about temperature or rainfall? Does yesterday's weather provide information beyond the season? How informative is a nearby station about today's local weather, and which differences remain when observations are summarised by month?

26. USGS earthquakes: event sizes, locations, and timing

Download. USGS Earthquake Catalog search. Results can be downloaded as CSV; the linked catalogue service also supports CSV output.

Earthquake records include event time, geographical coordinates, depth, and magnitude, with additional information about the event and its measurements. The search page provides access to events from a chosen place and period. The catalogue is maintained and updated by the United States Geological Survey.

Earthquakes vary greatly in location and size. Events can also occur in clusters, so a list of events in chronological order contains more than a collection of isolated magnitudes. Regions differ both in their seismic activity and in what can be recorded. The same number of recorded events can describe a geographically concentrated sequence or a widely dispersed one.

Questions. How does the distribution of recorded magnitudes differ between regions? How informative is location about depth? Does knowledge of recent events reduce uncertainty about the timing or location of subsequent recorded events, and how does this differ across periods?

27. NASA exoplanets: diversity beyond the Solar System

Download. NASA Planetary Systems Composite Data. The table's download menu provides CSV exports.

The archive compiles properties of confirmed exoplanets and their host stars. Available fields include planet radius, mass, orbital period, stellar properties, and discovery method. Some properties are missing, and many measurements have reported uncertainties or limits.

The observed planets span a wide range of sizes and orbits. Some properties may carry information about others, but the discovery method also affects which objects enter the catalogue. The data therefore bring together questions about the diversity of recorded planets and the information contained in incomplete or imprecise measurements.

Questions. How much information does radius provide about mass or orbital period? How informative are the host star's properties about a planet? Do different discovery methods reveal different distributions, and how much uncertainty remains when a property is represented only by a broad category?

10 Games and networks

28. StatsBomb football: passes, shots, and sequences of play

Download. StatsBomb Open Data repository. JSON files contain competitions, matches, lineups, and events for selected competitions and seasons.

The event records describe actions within a match, including passes and shots, their timing, and their locations. Player and team identifiers connect events within the same match. Some selected matches also include additional positional information.

Football contains recurring patterns but leaves many choices open at each moment. A team's use of space can be more or less varied, and a player's next action may depend on the preceding sequence. Event locations also provide different information from the identity of the player or team. Two teams can complete similar numbers of passes while using very different sequences and destinations.

Questions. How uncertain is the next action when the recent sequence is known? How informative is shot location about outcome? Do teams differ in the entropy of their passing destinations, and how much does player identity add beyond location and match situation?

29. Lichess: chess openings and choices over time

Download. Lichess open database. Standard rated games are supplied in monthly PGN files compressed with Zstandard, separate from chess variants.

Each game records its moves and result, together with fields such as player ratings and time control. PGN retains the order of play, so the position reached at any stage is related to the preceding moves. The monthly files provide fixed collections of online games.

Early play often follows recurring patterns, while later positions can offer a different range of choices. Players' ratings and the amount of thinking time may be related to both move choices and outcomes. Several games can begin identically and later diverge. A sequence can therefore be repetitive for a long stretch while leaving substantial uncertainty about the eventual result.

Questions. How much uncertainty about the next move remains after the opening moves are known? Do players of different ratings make equally varied choices? How much information do the opening and the players' ratings provide about the result, and how repetitive are the move sequences?

30. Facebook social circles: connections and overlapping groups

Download. Stanford SNAP Facebook dataset. Downloads include a compressed edge list and files for individual networks, circles, and anonymised features.

The combined graph has 4,039 nodes and 88,234 undirected edges. It was assembled from selected users' local Facebook networks. The accompanying files describe social circles for individual networks and features whose original meanings were anonymised.

Connections can reveal structure that a list of individual attributes does not show. People can belong to several circles, and some ties may repeat information already conveyed by other ties. A network can contain both highly connected regions and less connected ones. Two people with equally many connections can occupy different positions in this structure, with different relations to the recorded circles.

Questions. How much do a person's connections reveal about circle membership? When do several connections convey overlapping information? How much information about the network is lost when it is described only by each person's number of connections?